

AI Crypto Trading Journal

Self-hosted · Raspberry Pi 5 · Flask 3.1 · Claude Sonnet 4.6 + Haiku 4.5 · Bitget + Binance · SQLite WAL · 17 live data sources · Hermes Telegram agent

842	67.3%	+\$12,840	1.49	3.82
Positions	Win rate (Long)	P&L 90d USDT	Sharpe	Calmar

* Illustrative numbers only — not real results

AI Pipeline (per trade call)

Stage	Agent	Model / Source
1 Collect	agent_data_collector	14 parallel workers
2 Candles	chart_context	CCXT · Binance Futures
3 Interpret	chart_confluence / S&R	Pure Python indicators
4 Sentiment	market_context + Nansen	REST APIs
5 Review	backtest_context	SQLite trade history
6 Score	prompt_builder → Claude	Sonnet 4.6
7 Consensus	agent_orchestrator	Gemini Flash
8 Risk	agent_risk_mgmt	Kelly criterion + ATR
9 Chart	agent_chart_draw	matplotlib · S&R zones
10 Alert	telegram_notify	Telegram Bot API (push alerts)
11 Interact	Hermes agent	Interactive Telegram bot (pull queries)

Model Routing

Task	Model	Reason
call_analyzer	Sonnet	15-field JSON + chain-of-thought
scanner_batch	Sonnet	12 setups: entry/SL/TP reasoning
advisor	Sonnet	Full portfolio coaching
rulebook	Sonnet	Synthesises 800+ trade history
pattern_detect	Sonnet	Cross-pattern compounding
scanner_quick	Haiku	Score 0-10, runs 100× per scan
hindsight	Haiku	Binary ENTER/SKIP classification
live_trade	Haiku	Hold/Adjust/Close — latency key
trade_grader	Haiku	A–D grade, simple rubric

Claude + Gemini Consensus Scoring

[Δ] Claude – Gemini	Confidence	Score	Flag	Action
0 – 1 pt	High	Average	✓ Confirmed	Trade normally
2 pt	Medium	Average	~ Aligned	Trade, watch
3 pt	Low	Claude 60% + G 40%	■ Divergent	Reduce size
> 3 pt	Very Low	Claude kept	■ REVIEW	Do not trade — investigate

Data Sources — Macro → Micro

Layer	Source	Provider	Auth	Used in Pipeline
L1 Global Macro	VIX Volatility Index	CBOE · yfinance	Free	Scanner cap (>35→6.0, >25→7.5) · Confluence ×0.80 when >30
L1 Global Macro	DXY US Dollar Index	ICE · yfinance	Free	Macro regime label · Stage 3 scanner header
L1 Global Macro	Fear & Greed Index	alternative.me	Free	Scanner macro cap · Sentiment prompt · Dashboard strip
L1 Global Macro	Economic Calendar	Finnhub API	Key	FOMC/CPI/NFP → cap 7.0 when high-impact event in 24h
L1 Global Macro	ES1! S&P 500 Futures	yfinance (ES=F)	Free	Equity risk proxy · scanner macro header · risk-off signal
L1 Global Macro	BTC.D · USDT.D · OTHERS.D · TOTAL2/3	CoinGecko (free)	Free	Derived from market_cap_percentage · dominance indexes · alt season gauge
L1 Global Macro	MEME.C · STABLE.C · STABLE.COIN	CoinGecko /categories	Free	Sector caps · stablecoin dominance (falling=risk-on) · meme cap spike=top risk
L2 Market Struct.	Options Skew (PCR / IV)	Deribit (free)	Free	BTC/ETH only — put/call bias, near-term IV, sentiment label
L2 Market Struct.	BTC Mempool	blockchain.com	Free	On-chain congestion context injected into prompt
L2 Market Struct.	Trending Coins (top 10)	CoinGecko (free)	Free	Is the analyzed coin trending? Prompt context block
L3 Symbol-Level	Multi-Exchange L/S Ratio	Binance+Bybit+OKX	Free	Crowd positioning · contra-signal when >65% vs trade dir.
L3 Symbol-Level	OI · Funding · L/S · Liq trend	Coinalyze API	Key	Derivatives block · funding bias · per-exchange spread
L3 Symbol-Level	Historical Liquidations (daily)	liquidation_client	Key	longs_usd/shorts_usd per day · dominant side · CSV cache
L3 Symbol-Level	Cap rank · tier · volume	CoinGecko (free)	Free	Coin context · scales Grok weight (mega 30% → micro 80%)
L3 Symbol-Level	DeFi TVL + 7d change	DefiLlama (free)	Free	DeFi tokens only — protocol health context
L3 Symbol-Level	OHLCV Candles (4H + 1D)	Binance · CCXT	Free	All indicators · S/R detection · SMT · Backtester · Charts
L4 Trade Intel.	Smart Money Wallet Flows	Nansen API	Paid	Accumulating/distributing ■/■ in sentiment agent prompt
L4 Trade Intel.	Social & News Context	xAI Grok API	Key	Direction-weighted; last prompt block · lowest priority

What's New — v1.6.0

HTF→LTF Breakdown — 1H candles for entry/SL (fresher), 4H confirmation, 1D bias, 4H/1D for TPs — cuts staleness 4h→1h

Dominance Indexes — BTC.D · USDT.D · OTHERS.D · TOTAL2/3 · MEME.C · STABLE.C.D · ES1! — all in scanner header

S&R Chart Improvements — Green support / red resistance zones; ATR-based width; confluence merge; at-level ■ highlight

Hermes Telegram Agent — Interactive bot on Pi: scan coins, get P&L, price check, chart send, whale signals, dominances

Scanner Cancel Button — X Cancel stops running scan at next stage boundary — no DB writes, no alerts fired

Direction-Aware Analysis — Short position monitor now correctly treats bearish momentum as favorable; SL placed above entry

Pending Limit Fixes — Verdict display fixed; Bitget preset SL/TP auto-synced; scanner chart shown inline in card

17 Data Sources — Added ES1! (yfinance), USDT.D/TOTAL3/MEME.C (CoinGecko), liquidation history (Coinalyze)

Technical Stack

Layer	Technology
Runtime	Python 3.13 · Flask 3.1
Database	SQLite WAL · 7-day auto backup
AI Primary	Claude Sonnet 4.6 · Haiku 4.5
AI Consensus	Google Gemini 2.0 Flash
AI Social	xAI Grok 3 Fast
Exchange	CCXT · Binance · Bybit · OKX
Charts	mplfinance · lightweight-charts
Hardware	Raspberry Pi 5 · port 8082
Tests	pytest · 351+ passing

Prompt Budget Order

Priority Block		Protected until
1st	Backtest context	Always — most trade-relevant
2nd	Market data blocks	Always (Coinalyze, F&G, L/S, options...)
3rd	Rulebook	Remaining < 100 chars
4th	Chart context	Remaining < 100 chars
5th	Calibration	Remaining < 100 chars
6th	Grok social	Remaining < 150 chars · weight 0–80%